



The application of bioturbators for aquatic bioremediation: Review and meta-analysis[☆]

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ABSTRACT

Human activities introduce significant contamination into aquatic systems that impact biodiversity and ecosystem function. Many contaminants accumulate, and remediation options are now required worldwide. One method for bioremediation involves the application of macrofauna to stimulate microbial ecosystem processes including contaminant removal. However, if we are to confidently apply such a technique, we need clarity on the effect of bioturbators on different contaminants and how these vary under different environmental scenarios. Here we used a systematic review and meta-analysis to analyse current knowledge on the activities of bioturbating macrofauna in contaminated sediments and quantify how bioturbation-bioremediation changes depend on the taxonomic group, the aquatic ecosystem and important environmental variables. Three common contaminant classes were reviewed and analysed: metals, nutrients (i.e. ammonia and phosphorous) and polycyclic aromatic hydrocarbons (PAH). In addition, meta-regressions were calculated to estimate the effect of environmental and experimental design variables on effect sizes. Meta-analytic results revealed that deeper burrowing and more active sediment surface animals (e.g. polychaetes) increased metal release from sediments, nutrients and oxygen uptake by microbial fractions in comparison to bioturbators that inhabit shallower depths in sediments. In addition, there was a different effect of bioturbators on response variables in different aquatic systems. Finally, bioturbator effects on nutrient and metal release appeared modulated by context-specific variables such as temperature, pH, sediment grain size, animal density and experimental duration. Our findings highlight critical knowledge gaps such as field applications, less studied macrobenthic fauna and the incorporation of molecular approaches. Our results provide the first quantitative synthesis of the effects of bioturbators on contaminant fate and the variables that need to be considered for the optimization of this method as a viable approach for sediment remediation and contaminant management in aquatic systems.

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1. Introduction

Human activities have increased the number and concentrations of contaminants entering natural aquatic ecosystems (Chapman and Wang, 2001). Some of the most common contaminants include metals, hydrocarbon compounds and organic matter (Skei et al., 2000). These contaminants may have negative effects

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across multiple ecological levels and can interact with other environmental variables to cause synergistic impacts on aquatic ecosystem structure and function (Johnston et al., 2015; Johnston and Roberts, 2009). For example, metals can be sequestered into sediments, chelate with organic matter affecting biogeochemical cycles or be absorbed by local organisms and transferred through the food chain to higher trophic levels (Tchounwou et al., 2012). In addition, high concentrations of hydrocarbons can directly influence the mortality of key species and be transferred into or magnified at other trophic levels (Grall and Chauvaud, 2002). In both cases, the loss of important functional taxa can change nutrient dynamics in the system and lead to poor ecosystem health and eutrophication (Douglas et al., 2017). In these instances,

environmental remediation often requires human intervention to remove or decrease the effect of contaminants and restore ecological function at degraded sites (Khan et al., 2004). However, remediation applications can themselves be damaging (Boopathy, 2000) and environmentally friendly solutions are needed to address the global problem of contamination.

Physical remediation (e.g. dredging) and chemical remediation (e.g. sediment washing) are the most common methods of removing contamination in many ecosystems (Adriaens et al., 2006). However, biological remediation (bioremediation) of contaminated sites has become an increasingly popular alternative due to its potential for cost effectiveness and lower environmental impact (Vidali, 2001; Meuser, 2012). Bioremediation uses living organisms that interact directly or indirectly with the environment to neutralize or remove contaminants (Iwamoto and Nasu, 2001). Bioremediation can be applied both *in situ* (i.e. field conditions) and *ex situ* (i.e. mesocosm/controlled conditions). Many small-scale examples exist of both approaches using plants, fungi and bacteria as bioremediators of organic contaminants, although results have been varied (Vidali, 2001; Harms et al., 2011). However, to date there have been few applications of bioremediators at a large scale (Boopathy, 2000; Adriaens et al., 2006).

An emerging area of bioremediation research involves the application of bioturbating macrofauna. Bioturbation comprises a series of processes driven by macrobenthic fauna that affect sediment physical and chemical properties and strongly influence bacterial communities involved in nutrient cycling (Biles et al., 2002). Bioturbation considers both particle reworking, bioirrigation and other benthic biota behaviours (i.e. feeding or grazing) that are involved in the transport and movement of porewater and particles through the water-sediment interface (Kristensen et al., 2012). Sediment reworking and bioirrigation can affect the mobilisation of dissolved and particulate contaminants by increasing the transport and biomixing rate from overlying water and porewater to deep layers of the sediment (Timmermann et al., 2000; Teal et al., 2013). Bioturbators can also increase contaminant sequestration by the formation of complex biofilms derived from bioturbator organic waste (Lalonde et al., 2010; Qin et al., 2010). In addition, bioturbation activities can increase benthic metabolism and nutrient dynamics by the stimulation of aerobic bacterial communities that participate in contaminant degradation (Kunihiro et al., 2011; Ma et al., 2015; Boeker et al., 2016; Shen et al., 2017).

Our understanding of the effects of bioturbation on contaminant bioremediation is incomplete and the results appear to be inconsistent with a diverse range of responses reported with respect to different classes of contaminants, magnitude of contamination and geographical region (Maximov et al., 2015; Natálio et al., 2017). Further, there is a lack of information relating to the interaction of bioturbators and microbial communities, and the importance of this interaction for contaminant fate in sediments. With the increasing impact and widespread distribution of contaminants in aquatic systems, there is an urgent demand for remediation options and bioturbation may be an important alternative. A review and synthesis of the available literature is both timely and necessary to assess the potential for bioturbation-bioremediation as an alternative approach for contaminant removal. Information generated through this approach may provide a framework to optimize bioremediation efforts by defining which bioturbator groups have the greatest effect on different contaminants, and if their application in aquatic systems is sufficient to reduce contaminant bioavailability.

Here we used a systematic review and meta-analysis to assess the current state of knowledge of the effect of macrobenthic bioturbators on contaminant fate in aquatic sediments and how this is

influenced by the taxonomic group, the aquatic system and important abiotic/biotic environmental factors. In addition, this review reveals the frequency at which different bioturbation strategies have been taken to practice for decreasing sediment contamination and identifies knowledge gaps hindering the development of bioturbation as a viable method for bioremediation.

2. Methods

2.1. Search methodology

An initial literature search was done to identify the main contaminants in sedimentary systems that have had bioturbators applied for remediation. Search terms included bioturbation, bioremediation, pollutants, sediment reworking, bioirrigation and sediments. Searches were done in ProQuest, Web of Science and Scopus. From this search, three main contaminants were identified: metals, organic matter/nutrients and polycyclic aromatic hydrocarbons (Suppl. Fig.1).

Systematic searches were conducted to identify studies where the effect of bioturbation on each contaminant was evaluated. Search terms included those listed above and additionally: ecological engineer, macrobenthic, biological disturbance, remediation, microb* and bacter*. These terms were combined in searches with selected contaminant terms: metals (heavy metals, metals), organic matter and eutrophication (organic matter, organic enrichment, nutrient, eutrophic*) and polycyclic aromatic hydrocarbons (PAH, Polycyclic Aromatic Hydrocarbons, persistent organic pollutants). In total, 396 searches were conducted for all evaluated contaminants in each search database using all the pertinent combinations of the search terms. Searches were captured from March to August 2017. Duplicate articles were removed, and relevant articles were selected by reading titles and abstracts. Full details of the search processes and literature search are included in Supplementary Figs. 2–4 and Supplementary Table 1.

2.2. Qualitative analysis

In total, 222 research articles satisfied the criteria for inclusion in the review. From these, we found 56 articles relating to metals, 133 articles for organic matter and nutrients, and 33 articles relating to PAHs. From these, information about the main taxonomic classes, location of study, country and aquatic systems were collected as well as details of methodological approaches (*ex situ* or *in situ*). Aquatic system classification was obtained directly from each individual study in accordance with how authors described their experimental site. The number of articles where microbial analyses were used to assess the impact of bioturbation on bacterial and archaeal communities on contaminant fate was also reported. Finally, the number of cases where bioturbation was explicitly reported in a real-world bioremediation project was also documented.

2.3. Design criteria for meta-analysis

The purpose of the meta-analysis was to determine the effect of bioturbators on contaminant fate in sediments. From the initial 222 articles found, only 49 articles met the criteria to conduct a meta-analysis for metals (13 articles) and organic matter/nutrients (36 articles). For metals, insufficient data was available to separate particulate and dissolved fractions and therefore data from total metal concentrations in the water column were included as the specific variable for the analysis. For organic matter and nutrients,

three variables representing a broad range of effects of bioturbation on nutrient release were selected: ammonia (30 articles) and phosphorous fluxes (18 articles) from sediment surface to water column, and sediment oxygen uptake (SOU) (25 articles). The selected variables were investigated in the literature as end responses related to the presence of bioturbation and as indirect measurements of bacterial activity.

Total Ammonia ($\text{NH}_3 + \text{NH}_4^+$) was evaluated as a variable for the analysis as it is an important product of bacterial aerobic metabolism during nitrogen dynamics (Martinez-Garcia et al., 2015; Zhang et al., 2014). Total phosphorous (TP) and orthophosphate (PO_4^{3-}) concentrations in the water column were also included in the analysis as these molecules represent an important source of nutrients for algal growth and thus influence eutrophication (Renz and Forster, 2014; Chen et al., 2016). SOU was evaluated as an indirect measure of bacterial aerobic metabolism in the sediment (Renz and Forster, 2014; Martinez-Garcia et al., 2015). No meta-analysis was conducted for PAHs as there were not enough comparable results and methodologies.

2.4. Meta-analyses of the data

From all studies selected for each response variable (i.e. total metals, ammonia, phosphorous and SOU) the mean, standard deviation and number of replicates were extracted or calculated from the raw data directly from the scientific articles. Graph data were extracted using the online application WebPlotDigitizer 4.1 (Rohatgi, 2018). In addition, other relevant variables like temperature, pH, salinity, experimental duration, sediment size and animal density were extracted for inclusion in analyses as covariates involved in mediating the response variables. All studies that evaluated the presence or absence of soft sediment macrobenthic and epifaunal bioturbators were considered for this analysis.

For each selected response variable, a multi-level model (package metafor R, Viechtbauer, 2010) was used to account for non-independent sampling error due to correlations between effect sizes sharing the same control (Noble et al., 2017). The effect size for each study was calculated by the standardized mean difference (SMD) and sampling variances were obtained. Multiple SMD were obtained per study and effect size estimates were correlated through a variance-covariance matrix to account for non-independence. Absence and presence of bioturbators were selected as fixed factors for the model and animal class and system type were included as nested random factors. In all cases, family-wise error rate between multiple comparisons for each random factor was calculated using a Bonferroni correction. Forest plots were constructed for each response variable using the forest function (package metafor R, Viechtbauer, 2010; Supplementary Figs. 5–8) to visualize the individual effect size, sample variance of each study and overall effect size estimate of the model. To assess relationships between effect sizes and additional moderator variables obtained during data extraction, multiple meta-regressions were done using the function *mareg* from the R package *MAAd* (Del Re and Hoyt, 2014). In addition, a funnel plot and a trim and fill method (Duval and Tweedie, 2000) were constructed to assess for publication bias from each of the evaluated contaminant response variables (Supplementary Fig. 9).

3. Results

3.1. Qualitative assessment of contaminant types in sediments tested with bioremediation

In this systematic review most studies were reported from temperate areas (e.g. USA, Australia, and European countries) with

few from tropical regions (Supplementary Fig. 10). Across multiple aquatic systems, most studies were conducted in *ex situ* laboratory, indoor and outdoor conditions, although some have attempted *in situ* approaches (e.g. Shen et al., 2016). For metals and PAHs, the majority (~80–90%) of bioturbator applications were conducted *ex situ*. By contrast, organic matter/nutrients had a greater proportion of bioremediation studies applied *in situ* (>40%) but still fewer than *ex situ*. Finally, experiments with bioturbators and PAHs were rarely reported on *in situ* conditions (~9%).

Estuaries, rivers and lakes are the main focus of studies investigating the effect of bioturbation on contaminants. Artificial systems, such as ponds or aquaculture systems are also represented, although to a lesser extent in all selected contaminants (Fig. 1). In addition, studies of metal bioremediation utilized mainly Oligochaetes ($n = 25$, mainly Tubificidae) from freshwater systems, while polychaetes ($n = 18$, Arenicolidae and Nereididae families), Bivalves ($n = 15$, Tellinidae) and Malacostracans ($n = 14$, Corophiidae and Pontoporiidae) were used in estuarine systems (Fig. 2). For organic matter and nutrients, polychaetes represented the most commonly used organisms to evaluate the effect of OM/nutrient and bioturbation, mainly with families such as the Arenicolidae, Capitellidae and Nereididae ($n = 40$). In addition, Malacostracans ($n = 31$) and Bivalves ($n = 25$) were common bioturbators in OM studies (Fig. 2). PAH fate included mostly nereidids and capitellids from the class polychaeta ($n = 15$), followed by tubificid worms ($n = 7$, Oligochaeta) and several groups of Malacostracans as main groups of interest ($n = 7$) (Fig. 2).

3.2. Meta-analysis of bioturbator effects on selected contaminants

Overall effect size determines the mean magnitude of effect of the evaluated responses in all studies included in the analysis. Results show that bioturbators were effective stimulators of organic matter degradation, contributed to nutrient release into the water column but did not affect metal release from sediments (Ammonia and Phosphorous $p < 0.001$; metals $p = 0.510$; Fig. 3). Polychaetes, malacostracans, and oligochaetes increased ammonia and phosphorus fluxes from sediment (Fig. 4). Bivalves and insects acted similarly, but only for ammonia fluxes (Fig. 4; Suppl. Table 2). However, Priapulids and echinoderms (classes Ophiuroidea, Holothuroidea) did not affect nutrient release. Interestingly, the effect of bioturbators on ammonia and phosphorous release was restricted to marine, Stormwater Infiltration Systems (SWIS), lakes and estuaries (Fig. 5). In rivers and fjords, bioturbators increased ammonia release when present, but had no effect on phosphorous (Fig. 5). For streams, sea inlets, ponds, mussel farms, coastal lagoons and aquaculture ponds, no release of ammonia or phosphorous was observed (Fig. 5, Suppl. Table 3).

Oxygen uptake into sediments increased when bioturbators were present ($p < 0.001$, Fig. 3). Oligochaetes and insects increased oxygen uptake into the sediment when evaluated by class. By contrast, bivalves decreased SOU, while gastropods, priapulids, polychaetes, ophiuroideans and malacostraca didn't affect SOU (Fig. 4, Suppl. Table 2). Bioturbation effects on SOU were variable across different ecosystems. Bioturbators increased SOU in fjords, SWIS and lakes, but decreased SOU in rivers and estuaries (Fig. 5; Suppl. Table 3). For the rest of the aquatic systems evaluated, no effect of bioturbators was documented.

Overall, the presence of bioturbators did not affect metal release from sediments ($p = 0.510$; Fig. 3) although some taxa-specific effects were observed. Specifically, ray-finned fishes increased metal sequestration. In comparison, bivalves, insects and oligochaetes had no statistically significant effect on metal release to the water column, although insects tended to increase metal release from the sediments (Fig. 4, Suppl. Table 2).

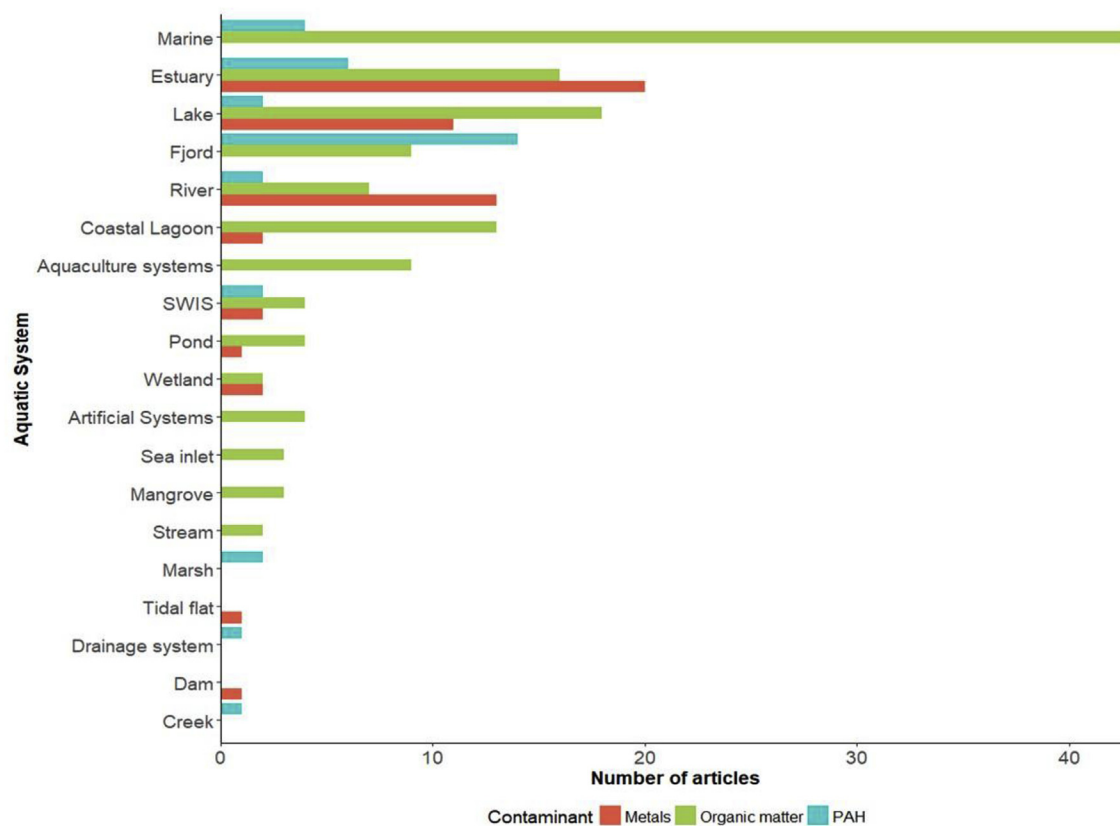


Fig. 1. Frequency diagram of main aquatic systems where the effect of bioturbation on contaminant fate were studied.

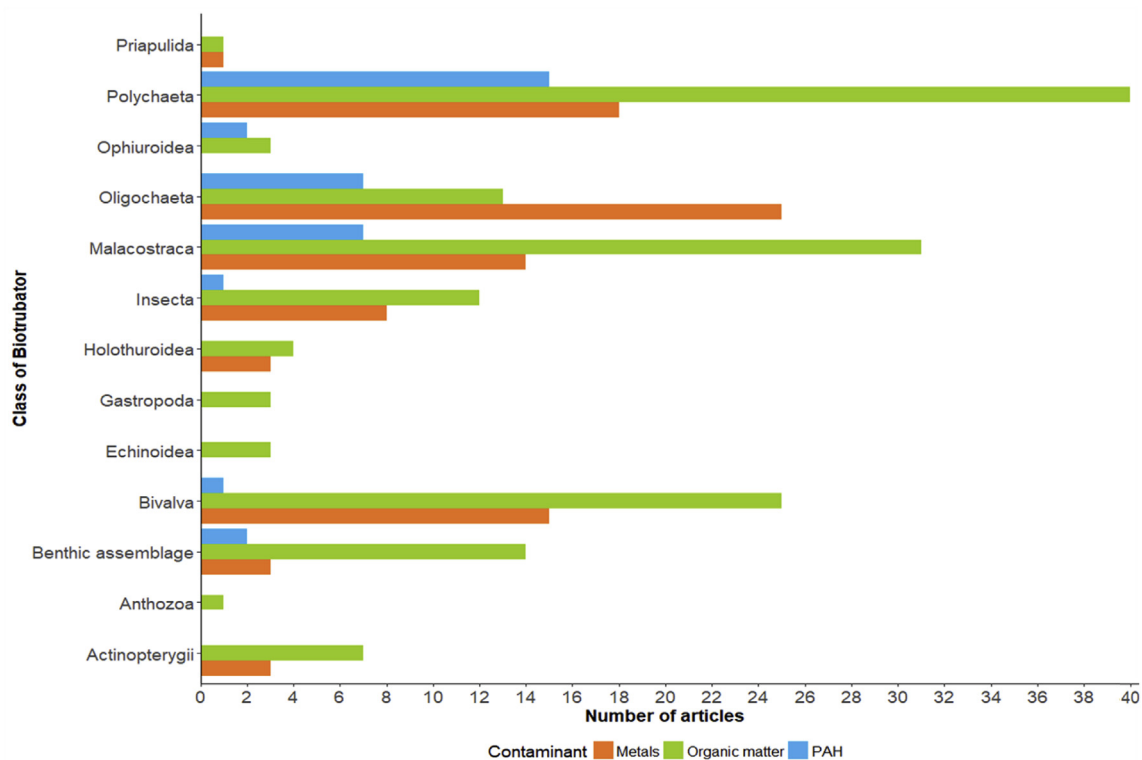


Fig. 2. Frequency diagram of the main taxonomic groups where the effect of bioturbation on contaminant fate were studied.

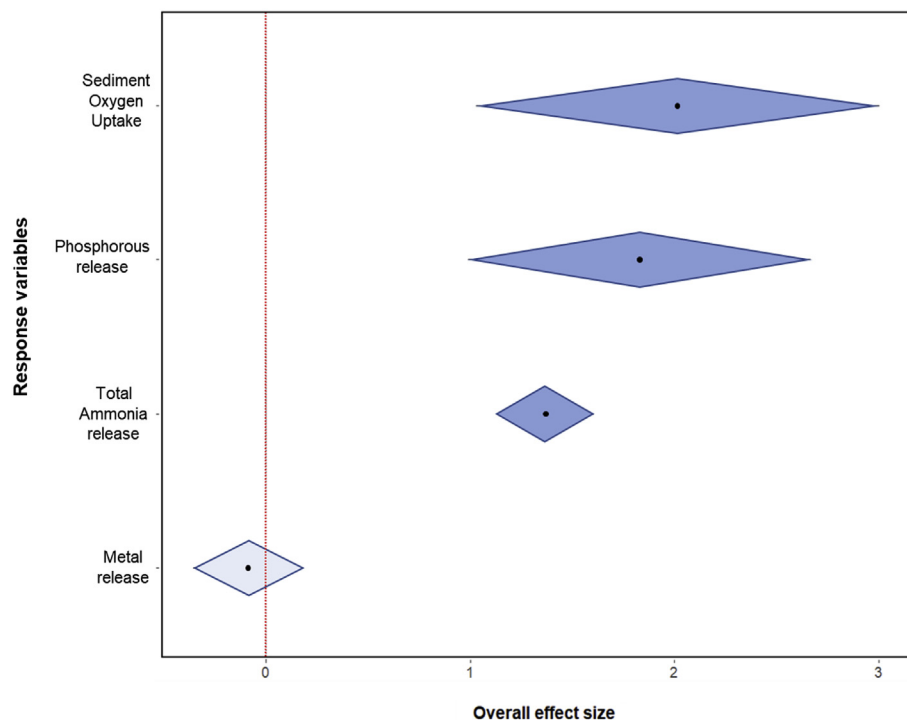


Fig. 3. Overall effect size of the release of evaluated responses from sediments to the overlying water and sediment oxygen uptake (SOU) in the presence of bioturbators. Point marks the overall effect size while polygon length represent overall confidence interval for all studies considered in each response. Significant effect sizes are represented by shaded polygons. Red line shows an effect size equal to 0. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

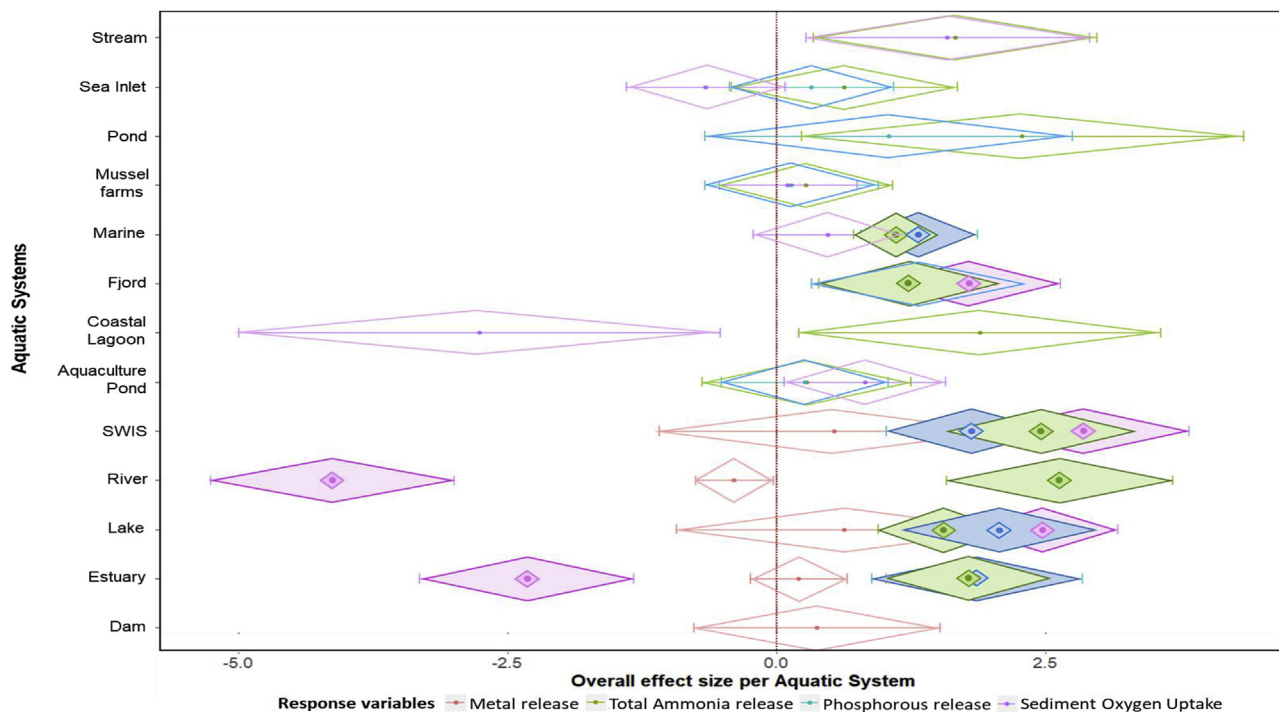


Fig. 4. Overall effect size of the evaluated responses in the presence of bioturbators and between different aquatic systems. Small polygons and points mark the overall effect size while large polygon length represents the confidence interval for all studies considered in each response. Significant effect sizes are represented by shaded polygons. Red line shows an effect size equal to 0. SWIS= Stormwater Infiltration System. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

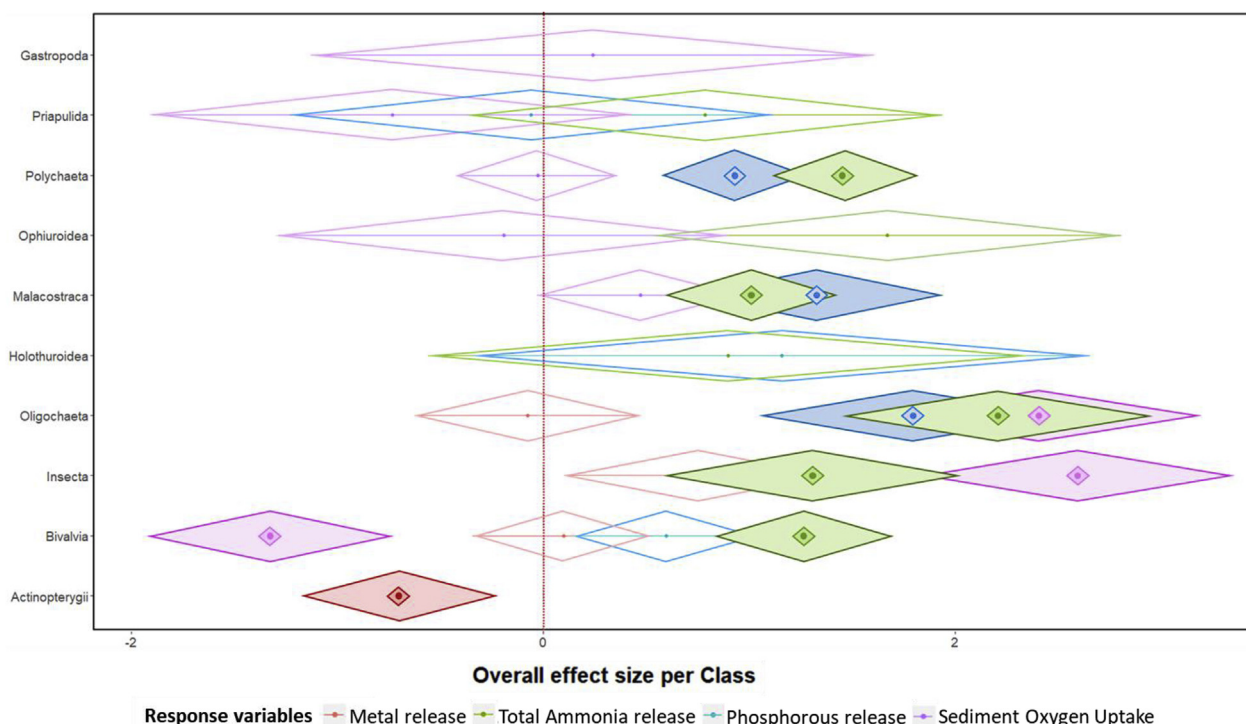


Fig. 5. Overall effect size of the release of evaluated contaminants from sediments to the overlying water and sediment oxygen uptake in the presence of bioturbators and between different taxonomic groups. Point marks the overall effect size while polygon length and height represent range of effect the effect size for all studies considered in each response. Significant effect sizes are represented by shaded polygons. Red line shows an effect size equal to 0. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3.3. Meta-regression analysis of effects of moderator variables

Meta-regression analysis highlighted moderator variables that changed the effect of bioturbators on contaminant release. For example, higher temperatures increased the release of ammonia and decreased phosphorus release when bioturbators were present (Suppl. Fig. 11). By contrast, metal release and SOU were not correlated to temperature when bioturbators were present, although there was a positive trend for metal release (Suppl. Fig. 11, $p = 0.110$). In addition, for metals, increasing pH levels show increased metal solubility from sediments in the presence of a bioturbator (Suppl. Fig. 11, $p < 0.001$). Salinity was not correlated to ammonia, phosphorous or SOU (Suppl. Fig. 11).

Sediment grain size was positively correlated with the release of ammonia, phosphorous and SOU, suggesting that bioturbators were more effective in coarser sediments (Suppl. Fig. 11). In another case, increasing animal density affected phosphorous release and SOU but did not affect sediment ammonia release (Suppl. Fig. 11). Not enough sediment grain size and animal density metadata was found for metals in order to conduct this test. Lastly, metal, ammonia and phosphorous release and SOU were greater overall for longer studies (Suppl. Fig. 11). All meta-regression results for all the response variables are shown in Supplementary Table 4.

Funnel plots and a trim and fill method indicate no publication bias for metal and phosphorous release or SOU (Suppl. Fig. 8). However, a possible publication bias was determined for ammonia release ($k = 6$, $p = 0.0078$). However, Funnel plot results could be associated to other sources of asymmetry, like differences in methodological quality between studies, heterogeneity of results or data irregularities (Egger et al., 1997).

4. Discussion

The legacy and continuing contamination of sediments in aquatic systems is a pressing global issue that requires novel and effective approaches for remediation. Here we synthesised current understanding of bioturbator effects on contaminated sediments and identified knowledge gaps to progress the research field. Overall, we found that bioturbators can influence nutrient dynamics by increasing organic matter breakdown and the release of ammonia and phosphorous to the water column. Sediment oxygen uptake, analysed as a measure of microbial aerobic activity, was also enhanced in the presence of bioturbators. By contrast, we found no overall effect of bioturbators on sediment metal release, but for one group of fishes which increased metal sequestration. There were fewer studies of metals and PAHs than organic matter, which did not allow for a robust meta-analysis of bioturbator effects on PAHs and may have influenced the findings for metals.

4.1. Bioturbators promote nutrient release and oxygen uptake

Eutrophication of the water column results from excess nutrients stimulating overgrowth of algae and the subsequent depletion of oxygen as algae die. Results from this review show that the activity of bioturbators, including polychaetes, malacostracans and oligochaetes can increase the release of both ammonia and phosphorus from sediments to the water column, while bivalves and insects stimulate the release of ammonia only. Many authors have attributed differences in nutrient release between taxa to differing bioturbation mechanisms (Michaud et al., 2006; Shen et al., 2016). For ammonia, increased fluxes are end products from aerobic microbial activity stimulated by the presence of a bioturbator. This

process is especially important in burrows which have been described as OM ‘hotspots’ where high levels of OM accumulation and degradation occur (Gerald et al., 2017; Poulsen et al., 2014; Shang et al., 2013; Holmer et al., 2015; Natálio et al., 2017). Phosphorous can also be stored in burrows, however the presence of bioturbators can increase its release through bioirrigation activities (Norkko et al., 2012; Ekeröth et al., 2012; Chen et al., 2016). Released phosphorous to the overlying water column by bioturbation can potentially become an important factor that could enhance eutrophication of the water column while simultaneously improving the ecology of the sediments (Sinha et al., 2017). It is therefore important to consider the fate of nutrients post-release before applying bioturbation to the problem of enriched sediments.

It is also possible that bioturbators that produce deeper biogenic structures can introduce higher amounts of oxygen and impact a larger surface area for nutrient release in comparison to surficial bioturbators (Jordan et al., 2009). This could in turn stimulate aerobic microbial activity that changes nutrient dynamics (Poulsen et al., 2014; Shen et al., 2017). No data could be obtained for polychaetes or malacostraca in the case of SOU, yet the results comparing oligochaetes and insects demonstrate that aerobic microbial metabolism is increased in the presence of the oligochaetes, an active bioturbator. Bivalves increase sediment oxygen release in comparison to other bioturbators, such as polychaetes or echinoderms (e.g. Rao et al., 2014; Lindqvist et al., 2009). This could be related to intermittent bioirrigation caused by bivalve surficial bioturbation mechanisms, as compared to more periodic oxygen fluxes of other infauna (Michaud et al., 2005; Volkenborn et al., 2012). These findings suggest a differential role of surficial bioturbators in nutrient dynamics and an important example of functional bioturbation groups in benthic organic matter degradation.

4.2. Bioturbator effects on metal remediation limited to one taxonomic group

The effect of bioturbation on metal fate in sediments is poorly represented in the literature, but there is evidence that epifaunal organisms such as fish may promote a higher metal sequestration in sediments than other functional groups of macrobenthic fauna (Pedro et al., 2015). Metal sequestration or bioaccumulation is favoured by sediment reworking and high bioirrigation rates from bioturbators, but this could only be evaluated in one taxonomic group in this study. Rayed finned fish appear to enhance metal sequestration in the sediment, however results should be interpreted with caution as a low number of studies were obtained for the present meta-analysis. Epifaunal bioturbators, such as rayed finned fish, have a higher motility and cover larger distances in less time that could account for increased sediment surface resuspension and metal binding with particulate material in comparison to other bioturbators (He et al., 2015; Wall et al., 1996). Rayed finned fishes have been reported to decrease metal release through sediment reworking and bioirrigation to enhance metal complexation in sediment surface (Wall et al., 1996; Pedro et al., 2015), however the present work lacks enough power to confirm this and most studies point to the opposite. Under this scenario, it is probable that rayed finned fishes are increasing metal release but at the same time enhancing accumulation through pellet formation that would reduce metal concentrations in the water column (Hodson, 1988; Pedro et al., 2015). Insecta also presented a pattern of effect on metal release, however this was not statistically significant. In this case, insects cause a lower accumulation of metals in sediments than other macrobenthic and epibenthic fauna mainly because of their different mechanisms of feeding and reworking the sediment (Colombo et al., 2016).

4.3. Further research on bioturbation effect on PAHs fate in sediments

Polychaetes are by far the most studied taxa in relation to PAH contamination largely because they are tolerant to PAHs and have the capacity to biotransform these compounds and reduce their toxicity (Christensen et al., 2002). Nonetheless, in other taxonomic groups PAH toxicity can affect sediment reworking and bioirrigation rates that in turn influence PAH mobilisation and bioavailability to other trophic levels (Timmermann et al., 2000). In the present study, however, not enough articles with comparable methodologies or enough information to develop a meta-analytic approach were found. Further investigations are needed in order to quantify adequately the impact of bioturbators on PAH fate and their importance as stimulators of aerobic microbial fractions that could enhance PAH degradation.

4.4. Factors that modulate the effect of bioturbators

The effect of modulator variables on bioturbators and the release of contaminants from sediments is an important issue to account for if bioremediation strategies are to be successfully employed. The heterogeneity of results obtained in the literature depend greatly on the context of abiotic and biotic variables, as well as different experimental designs and approaches.

Salinities vary in aquatic systems both naturally and because of human activities. This can change the rate of microbial activities and concordantly the release of nutrients and metals (Nowicki, 1994; Rysgaard et al., 1999). In our study, no clear patterns were found between higher salinity environments (i.e. marine) and lower salinity (i.e. freshwater systems) in relation to nutrient release or SOU (Suppl. Fig. 11). As each study was conducted in a specific site and different bioturbators with definite salinity tolerance thresholds and constant salinity conditions, no direct tests could be done to determine if salinity is a release modulating factor in the presence of bioturbators. However, results do demonstrate that in aquatic systems characterised by higher salinity, there is higher variation in ammonia release and SOU in comparison to other nutrients such as phosphorous, possibly as a result of other modulating factors (discussed below).

Meta-analytic results indicate that the effect of bioturbators on contaminant remediation varies among aquatic systems. Bioturbators tested in Stormwater Infiltration Systems (SWIS), lakes and estuaries affected ammonia and phosphorous release as well as SOU. Bioturbators tested in marine environments presented only an effect in nutrient release (i.e. ammonia and phosphorous) while rivers and fjords only affected ammonia release and SOU. For metals, no clear effects were seen and for PAHs most studies were associated to systems where PAH contamination is increased as a result of accidental release of oil (Boehm, 2005). Overall, this pattern may be linked to the dynamism of each system. The systems where an effect was observed, experienced greater water movement in comparison to more protected areas such as ponds, artificial systems and lagoons (Hasanudin et al., 2004).

There is a clear bias of studies being done in temperate regions such as parts of China, Denmark, USA, Germany and France for the analysed contaminants. Only 9% of reviewed articles assessed the effect of bioturbation on OM/nutrient fate in tropical regions, 4% for metals and none for PAHs. For polar regions, metals registered 4%, OM/nutrient 2% and 6% for PAHs. In both regions temperature is a key factor and meta-regression results reveal that temperature had a significant modulating role over the response variables for OM degradation and metals. Increased temperature produces an increase of ammonia release from sediments in response to the presence of bioturbation, likely due to higher bioturbation rates

resulting from higher metabolism of macrobenthic organisms and increased aerobic microbial activity in the sediment (Berkenbusch and Rowden, 1999; Bernard et al., 2016). However, our SOU results do not support this explanation as no significant correlation was found with temperature increase in this study. For other nutrients like phosphorus, an increase in temperature was correlated with a decreased flux from the sediment. In the literature, it has been reported that increasing bioturbation rate and higher infauna metabolism increased phosphorous release (Norkko et al., 2012; Ekeröth et al., 2012), the opposite pattern to what was found here. For metals, no significant correlation was found in this study, however it is possible that higher temperatures can be involved in an increase of metal resuspension and solubility as a consequence of higher bioturbation rates (Amato et al., 2016; Andres et al., 1998). Similar results are expected for PAHs, however the lack of information in the literature makes it hard to determine a clear effect of bioturbation and temperature on this contaminant in this study.

In the present study animal density, sediment grain size and pH were also correlated to increased contaminant release and aerobic microbial activity. Increasing bioturbator density increased the release of phosphorous and stimulated aerobic microbial activity (i.e. SOU) but did not increase ammonia release. It is possible that aerobic microbial activity is stimulated but does not enhance end products like ammonia, but instead increases other steps in OM mineralization (Nogaro and Burgin, 2014). Animal density is a crucial aspect for further investigation as higher densities may contribute to higher OM degradation but may also increase nutrients in the overlying water which can have counterproductive effects for remediation, depending on the ecosystem studied. For metals, abiotic variables like pH may be an important modulator variable (Suppl. Fig. 11) that determines different effects of bioturbation in multiple aquatic systems as it can change metal speciation and reactivity (Amato et al., 2016).

Sediment grain size is another important factor that determines a high degree of accumulation of OM and metals and changes the structure of microbial communities and metabolism (Jackson and Weeks, 2008; Burton and Johnson, 2010). In the current analysis, sediment grain size was positively correlated to ammonia, phosphorous release and SOU. Martinez-Garcia et al. determined that sediment grain size in the presence of a bioturbator was not significant at low levels of organic enrichment but in higher enrichments coarser sediments retain less OM and nutrients and microbial activity is increased (2015). This could explain the pattern seen in the present study and would be an indicator that the analysed studies in the meta-regression could probably derive from aquatic systems with high organic matter levels.

4.5. Future research directions

From the qualitative literature review, 28% of the analysed studies on nutrients directly proposed the application of bioturbators to remediate eutrophic sediments, while only 5% and 15% of studies focused on strategies for metal and PAH sediment remediation, respectively. OM/nutrient bioremediation through bioturbation has mainly been reported in artificial aquatic systems such as aquaculture ponds, mussel farms, enclosed artificial lagoons and recirculation aquaculture systems (Kang et al., 2017; Zhong et al., 2015; Bartoli et al., 2001; Nicholaus and Zheng, 2014; Kunihiro et al., 2008; Robinson et al., 2018; Lukwambe et al., 2018; Zhao et al., 2019). In these studies, water quality improvement of the aquaculture industry was the main focus to increase annual production rates (Biswas et al., 2009; Zhang et al., 2014; Kunihiro et al., 2011; Nogaro and Burgin, 2014; Shang et al., 2013; Holmer

et al., 2015; Robinson et al., 2015; Ren et al., 2010). Others have also proposed the use of bioturbation in bioreactors for the treatment of urban aquatic systems in order to treat municipal discharges (Penha-Lopes et al., 2009; Mermillod-Blondin et al., 2005, 2008). Nobbs et al. (1997) and Mayor et al. (2013) highlighted the importance of bioturbators (bivalves and malacostraca) as a component that will determine lake and estuary metal contamination from industrial wastewater. In addition, a similar proposal in stormwater infiltration systems as bioreactors has demonstrated the capacity of tubificids to immobilize metal release and even be applied in conjunction with other current bioremediation methods like phytoremediation (Nogaro et al., 2007; Hoang et al., 2018). Finally, for PAH contamination bioturbation-bioremediation was poorly represented in the literature as no study was directly linked to a bioremediation proposition for the exception of Kang et al. who evaluated the impact of mussels in PAH sediment removal in constructed wetlands (2019). For all analysed contaminants, there is a clear lack of reference to basic ecological theory, however results obtained in this study highlight relevant taxonomic groups and specific aquatic systems where sediment bioremediation should be effectively explored. Such is the case of artificial or urban aquatic systems in which development of a general bioturbation-bioremediation approach is still too early to apply but needs to be considered as an important tool for plausible solutions for sediment remediation in highly modified environments (Clark et al., 2015).

The present study highlights the lack of experimental data that approaches in more detail sediment macro-microbiota interactions. From the articles reviewed in this work, 25% revealed some information regarding the effect of bioturbation on the microbial communities involved in OM degradation. Some of these include studies done in recent years to establish the importance of both biological fractions for sediment bioremediation in aquaculture wastewater management (Robinson et al., 2018; Lukwambe et al., 2018; Zhao et al., 2019). The interaction between microbial communities present in the sediment and bioturbation on metals has also been poorly studied in the literature. From this systematic review, only 25% of the articles mentioned the microbial community interacting with metals and only Mayor et al. (2013) analysed directly a microbial quantifiable result (i.e. energy budgets) that can relate to the toxic effects of metals. Lastly, for PAHs, 27% of the reviewed literature analysed the effect of microbial communities stimulated by bioturbators for PAH degradation. Authors highlighted the importance of bioturbators to stimulate hydrocarbonoclastic microbial communities as a key factor for PAH degradation (Ciutat et al., 2006). In all analysed contaminants, bioturbation changed the physicochemical conditions of sediments, providing microbial communities with a higher complexity of microniches, higher resources and adequate redox environments to alter their abundance and structure (e.g. Boeker et al., 2016; Foshtomi et al., 2015). However, the lack of next generation molecular approaches has hindered the elucidation of the microbial mechanisms that can be a focus of interest to understand underlying processes involved in ecosystem function. The onset of new molecular approaches allows the possibility of having larger resolution and scale to analyse the mechanisms between macrobenthic biota and microbial communities in contaminated environments (Dafforn et al., 2014; Sun et al., 2012; Lawes et al., 2017). Results in this review show that bioturbation affects microbial sediment processes evidenced by SOU, however the study of macro-microbiota interactions has not been addressed in depth and new technologies make this possible and necessary to evaluate.

The majority of studies in this review were conducted under *ex situ* experimental conditions and in short time periods. OM/nutrient bioturbation presented 60% *ex situ* experiments. For

metals, 18% of the studies were approached *in situ*, where studies focused on the mechanisms of metal mobilisation and biomixing aided by changes in the redox state of sediments (Teal et al., 2013; Klerks et al., 2007). Only 10% of the extracted articles applied *in situ* experimental designs in order to assess PAH bioturbation-bioremediation strategies (Qin et al., 2010; Timmermann et al., 2011). In most *in situ* studies, application of bioturbators was done in small experimental areas (Gerald et al., 2017; Wang et al., 2010; Gammal et al., 2017) and only one was attempted at a larger scale (Ma et al., 2015). The response variables analysed generally increased over time, having the highest levels of release at 50 days from addition of bioturbators for phosphorous, metals and SO₄. However, ammonia release from the sediment decreased as experiment duration increased. It is possible that available OM in experimental mesocosms might start depleting and so ammonia release also decreases. This is a case which probably would not happen in a real-world field situation although it is likely in any experimental arrangement in the laboratory or field (Hill et al., 2011). Short duration studies and *ex situ* studies may not represent adequately the effect of bioturbation on contaminant fate in sediments for a bioremediation scenario to be assessed. In most evaluated studies, *ex situ* conditions cannot reproduce natural conditions entirely and most modulating factors are controlled (Burton and Johnson, 2010). The present work highlights the importance of these modulating variables and together with larger spatiotemporal field based experimental designs, the applicability of bioturbation as a bioremediation method can be adequately analysed in future research.

5. Conclusion

This synthesis of current studies on bioturbation highlights the possibilities of this approach for contaminant remediation. In the cases analysed in this review, bioturbation had a significant effect on contaminants through degrading, accumulating or decreasing bioavailability for other trophic levels. Few studies considered or differentiated the microbial mechanisms that are driven by bioturbating macrofauna and thus necessary for this remediation method to be effective. Context specific variables make it necessary to optimize this approach for regional ecosystems (Johnston and Keough, 2005). Overall, future research directions should aim to increase the number of field or semi-field experiments that concurrently measure macro scale responses. For example, changes in bioturbator performance and toxicity that impact sediment properties (i.e. sediment redox oscillations, contaminant dynamics) and cross-scale interactions (Burton and Johnson, 2010). In addition, the multiple environmental variables that have been analysed need to be considered as important modulator variables in order to optimize contaminant bioremediation. This will enable the creation of adequate strategies at longer spatiotemporal scales and assess the possibility of synergistic or antagonistic effects of multiple contaminants. The bioturbation-led remediation of contaminated aquatic habitats is possible, however, with our current understanding there is a need to develop site specific approaches using an adaptive management approach.

Authors' contributions

SVG conceived the idea and designed the methodology to use for the systematic review and meta-analysis; SVG collected and analysed the meta-data; SVG and KD led the writing of the manuscript. All authors contributed critically to the drafts and gave final approval for publication.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envpol.2019.04.023>.

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